

Multiple Choice (10 marks)

1. How many ways are there to arrange 6 pairs of parentheses such that they are balanced?
 - (a) 42
 - (b) 144
 - (c) 156
 - (d) 132
 - (e) 96
2. For any positive integer n , let $\langle n \rangle$ denote the closest integer to $\sqrt{n}$. Evaluate $\sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n}$.
 - (a) 7.0
 - (b) 2.0
 - (c) 4.5
 - (d) 1.0
 - (e) 5.0
3. If x and y are directly proportional and $x = 3$ when $y = 8$, what is the value of x when $y = 13$?
 - (a) 2.0
 - (b) 3.25
 - (c) 34.667
 - (d) 7.5
 - (e) 6.125
4. What is the number of equivalent parameter settings due to interchange symmetries in a mixture model with 10 components?
 - (a) 12000000
 - (b) 5000000
 - (c) 2000000
 - (d) 40320
 - (e) 7257600
5. What is the ones digit of $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9$?
 - (a) 9

- (b) 8
- (c) 0
- (d) 1
- (e) 6

True / False (10 marks)

Answer True or False

6. True or False: The eigenvalues of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ are 4 and 1.
7. True or False: The cost of the used car Manuel bought is represented as the numeral 6049.
8. True or False: Molly's final cost after a 15% tip on a \$28.75 haircut is \$28.75.
9. True or False: The correct answer to 'Express $0.\overline{17}$ as a common fraction.' is ' $\frac{8}{45}$ '.
10. True or False: The number 240 is a powerful integer because it has prime factors whose squares also divide 240.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In the geoboard shown, the points are evenly spaced vertically and horizontally. Segment AB is drawn using two points, as shown. Point C is to be chosen from the remaining 23 points. How many of these 23 points will result in triangle ABC being isosceles?

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[asy] draw((0,0)--(0,6)--(6,6)--(6,0)--cycle,linewidth(1));
for(int i=1;i<6;++i) {for(int j=1;j<6;++j) {dot((i,j));}}
draw((2,2)--(4,2),linewidth(1));
label("A",(2,2),SW);
label("B",(4,2),SE);
[/asy]
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12. Given a rectangle with dimensions of length $3x$ inches and width $x + 5$ inches, derive the value of x such that the area equals the perimeter, and explain your reasoning.

End of Paper